

Type 1+2 PV Multipolar Surge Protector

FXPV-12.5B series



Class I • Class II • Type 1 • Type 2

- Location of Use: String box, Inverter
- Mode of Protection: (DC+) - PE, (DC-) - PE, (DC+) - (DC-)
- Surge Ratings: $I_{total} = 12.5 \text{ kA}$ (10/350 μs)
 $I_{total} = 40 \text{ kA}$ (8/20 μs)
- IEC/EN/UL Category: Class I+II / Type 1+2
- Protective Elements: High Energy MOV
- Housing: Pluggable Design
- Compliance: IEC 61643-31
EN 61643-31
UL 1449 4th Edition

Technical Data

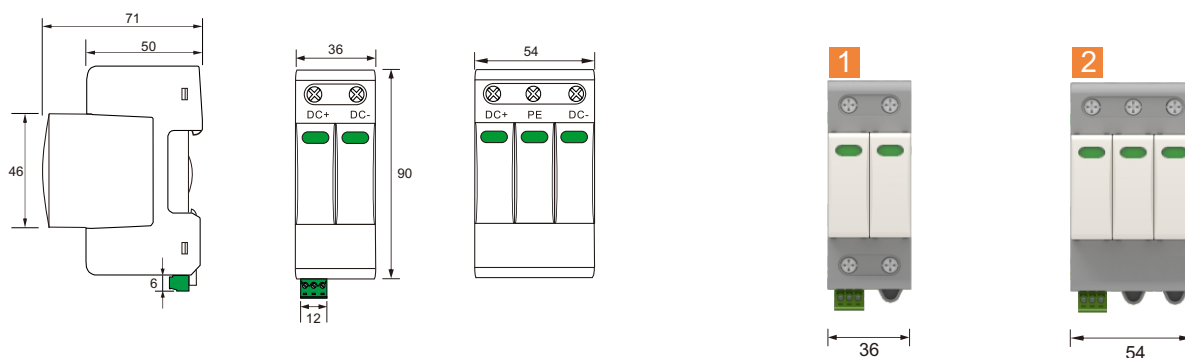
FXPV-12.5B series		600	1000	1200	1500	
IEC Electrical						
Maximum Continuous Operating DC Voltage	(DC+) - PE, (DC-) - PE (DC+) - (DC-)	U_{CPV} U_{CPV}	600 V	1000 V	1200 V	1500 V
Nominal Discharge Current (8/20μs)		I_n		20kA		
Impulse Discharge Current (10/350 μs)		I_{imp}		6.25kA		
Total Discharge Current (10/350 μs)		I_{Total}		12.5kA		
Total Discharge Current (8/20 μs)		I_{Total}		40kA		
Maximum Discharge Current (8/20 μs)		I_{max}		40kA		
Voltage Protection Level	(DC+) - PE, (DC-) - PE (DC+) - (DC-)	U_p U_p	≤2200V	≤4000V	≤4400V	≤5200V
Response Time		t_A		< 25 ns		
Short-Circuit Current Rating		I_{SCPV}		2000A		
Number of Ports				1		
Mechanical & Environmental						
Operating Temperature Range	T_a	-40 °F to +158 °F [-40 °C to +70 °C]				
Permissible Operating Humidity	RH	5%...95%				
Atmospheric pressure and altitude		80k Pa ... 106k Pa / -500 m ... 2000 m				
Terminal Screw Torque	M_{max}	39.9 lbf·in [4.5 Nm]				
Conductor Cross Section (max)		2 AWG (Solid, Stranded)/ 4 AWG (Flexible) 35 mm ² (Solid, Stranded)/ 25 mm ² (Flexible)				
Mounting		35 mm DIN Rail, EN 60715				
Degree of Protection		IP 20 (built-in)				
Housing Material		Thermoplastic: Extinguishing Degree UL 94 V-0				
Thermal Protection		Yes				
Operating State / Fault Indication		Green ok / Red defect				
Remote Contacts (RC)		Option				
RC Switching Capacity		AC: 250V / 0.5 A; DC: 250V / 0.1 A; 125 V / 0.2 A; 75 V / 0.5 A				
RC Conductor Cross Section (max)		16AWG (Solid)/ 1.5 mm ² (Solid)				

Type 1+2 PV Multipolar Surge Protector

FXPV-12.5B series

Dimensions - Internal Configuration

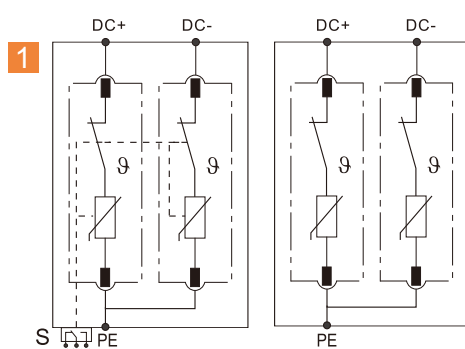
(in mm)



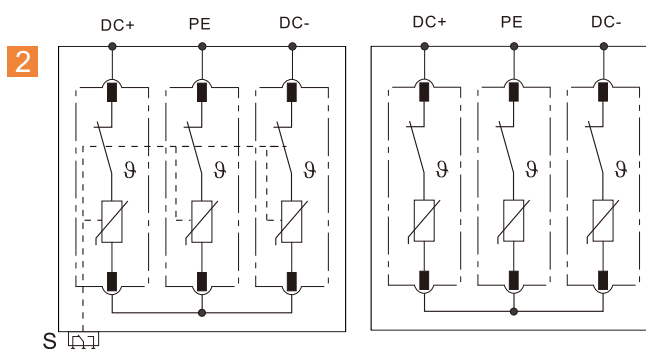
Legend

PE Protective Earth

S Signaling Contacts (Option)

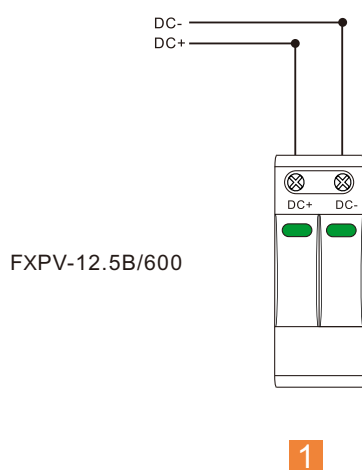


FXPV-12.5B/600

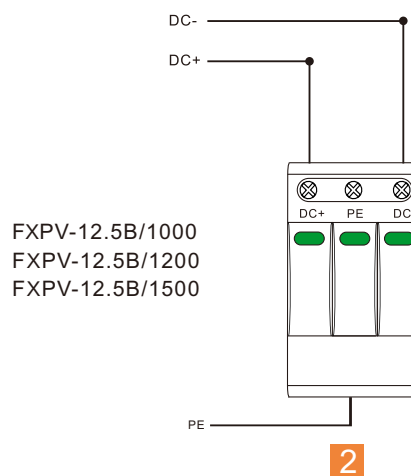


FXPV-12.5B/1000, FXPV-12.5B/1200, FXPV-12.5B/1500

Connection Diagram



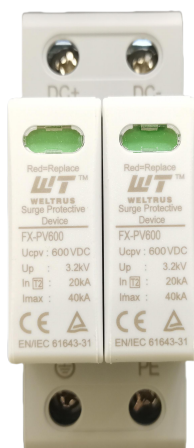
FXPV-12.5B/600



FXPV-12.5B/1000
FXPV-12.5B/1200
FXPV-12.5B/1500

Type 2 PV Multipolar Surge Protector

FXPV-XXX series



Class II...Type 2

- Location of Use: String box, Inverter
- Mode of Protection: (DC+)- PE, (DC-)- PE, (DC+)- (DC-)
- Surge Ratings: $I_n = 20 \text{ kA (8/20 } \mu\text{s)}$
 $I_{\text{total}} = 40 \text{ kA (8/20 } \mu\text{s)}$
- IEC/EN/UL Category: Class II / Type 2
- Protective Elements: High Energy MOV
- Housing: Pluggable Design
- Compliance: IEC 61643-31
EN 61643-31
UL 1449 4th Edition

Technical Data

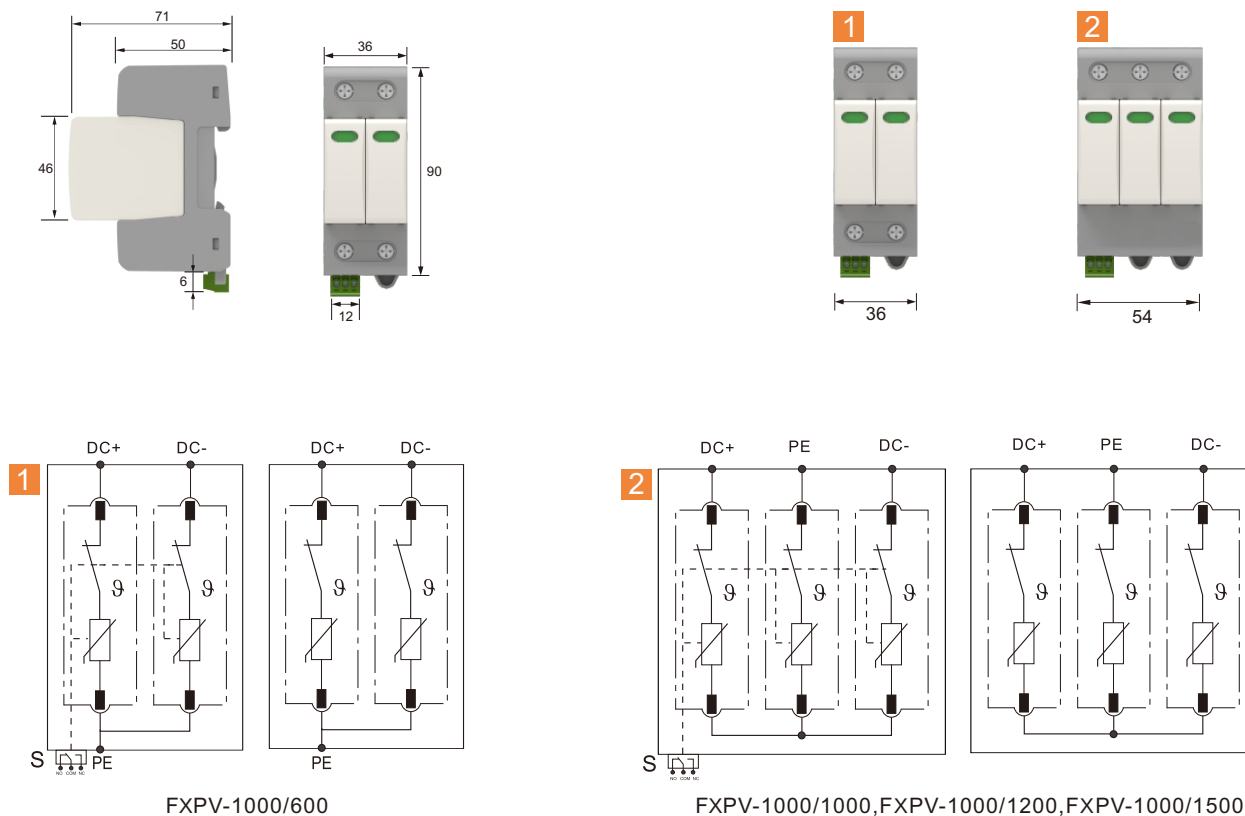
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IEC Electrical						
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Nominal Discharge Current (8/20 μs)		I_n		20kA		
Total Discharge Current (8/20 μs)		I_{Total}		40kA		
Maximum Discharge Current (8/20 μs)		I_{max}		40kA		
Voltage Protection Level	(DC+) - PE, (DC-) - PE (DC+) - (DC-)	U_p	$\leq 2200\text{V}$	$\leq 4000\text{V}$	$\leq 4400\text{V}$	$\leq 5200\text{V}$
Response Time		t_a		< 25 ns		
Short-Circuit Current Rating		I_{SCPV}		2000A		
Number of Ports				1		
Mechanical & Environmental						
Operating Temperature Range	T_a	-40 °F to +158 °F [-40 °C to +70 °C]				
Permissible Operating Humidity	RH	5%...95%				
Atmospheric pressure and altitude		80k Pa ... 106k Pa / -500 m ... 2000 m				
Terminal Screw Torque	M_{max}	39.9 lbf·in [4.5 Nm]				
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Type 2 PV Multipolar Surge Protector

FXPV-XXX series

Dimensions - Internal Configuration

(in mm)



Legend

PE Protective Earth

S Signaling Contacts (Option)

Connection Diagram

